

# Ten Layers Toward the Physical Riesz Interface

## A Conditional Closure Theorem, Rank-Growth Correction, and Minimal Frontier

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### Abstract

RH-161 identified the physical packet-to-Riesz bridge  $R$  as the first open operator interface inside Gate A. RH-162–170 now resolve its abstract proof architecture. We assemble those results into one conditional closure theorem with four physical leaves:  $X$ , a canonical ambient realization of the reset packet;  $D$ , validated finite resolvent and coupling data;  $K$ , uniform Schur and directional graph margins; and  $H$ , shellwise common-coordinate summable transport. If  $X$ ,  $D$ ,  $K$ , and  $H$  hold, every fixed reset shell lifts to a same-rank Riesz shell, the graph maps remain controlled, and the shells form a coherent all-level atlas.

The shellwise conclusion is necessary: RH-170 proves that full cloud projections of changing rank remain at least unit distance apart and cannot converge in operator norm. Relative to the displayed architecture, the unique minimal physical completion bundle is

$$\{X_{\text{phys}}, D_{\text{phys}}, K_{\text{phys}}, H_{\text{phys}}\}.$$

An aggregate audit rechecks 3,584 finite matrix cases and 63 exact rank-change witnesses with zero recorded failures. These validate formulas only. None of the four physical leaves is proved, so  $R$  and macro Gate A remain open. The first logically executable target is  $X_{\text{phys}}$ .

## 1 What changed after RH-161

RH-161 starts with a packet projection and a transfer operator on one Hilbert space [Program, 2026b]. The archived reset packet, however, is selected from a source-memory Gram [Program, 2026a], while the moving determinant cloud belongs to a noisy transfer operator [Program, 2026l]. RH-162 exposes the missing ambient realization rather than identifying equal-rank spaces by notation [Program, 2026c].

Once a realized packet exists, RH-163 replaces the symmetric Neumann gate by the directed product  $adbc < 1$  [Program, 2026d]. RH-164 gives its optimally balanced one-norm fallback and records the similarity condition penalty [Program, 2026e]. RH-165 solves the contour geometry for normal disk-separated blocks but explicitly rejects spectral-distance substitution for nonnormal blocks [Program, 2026f, Trefethen and Embree, 2005].

RH-166 identifies  $b, c$  as left and right Ritz residuals and separates primal from dual graph slopes [Program, 2026g]. RH-167 converts finitely many sample inverses into a continuous contour envelope, while RH-168 transfers that envelope through an operator ball [Program, 2026h,i]. RH-169 supplies common-coordinate fixed-rank transport, and RH-170 replaces the impossible global norm limit of a growing cloud by a shell atlas [Program, 2026j,k].

## 2 Four physical leaves

The proved formulas do not supply their own physical data. We collect the remaining inputs into four leaves.

**$X_{\text{phys}}$ : ambient realization.** For each scale  $j$  and shell  $m$ , construct a target-independent isometry  $J_{j,m}$  from the reset-memory space into the transfer/determinant space. Its polar normalization, source-dynamics commutator, primal defect, and adjoint defect are bounded. The realized packet is  $P_{j,m} = J_{j,m} P_{j,m}^{\text{preset}} J_{j,m}^*$ .

**$D_{\text{phys}}$ : validated finite data.** Choose a finite contour mesh  $\Gamma_{j,m}$  and provide outward bounds for nominal inverse defects, block operator balls, and both bi-Ritz residuals. RH-167–168 then produce exact continuous values  $a_{j,m}, d_{j,m}, b_{j,m}, c_{j,m}$ .

**$K_{\text{phys}}$ : uniform margins.** There are constants  $\kappa_* < 1$  and  $\delta_* < 1$  such that

$$a_{j,m} d_{j,m} b_{j,m} c_{j,m} \leq \kappa_*,$$

and the relevant primal, and when needed dual, packet-diagonal and graph bounds stay below their invertibility thresholds. A sequence of positive but vanishing margins is insufficient for a uniform assembly.

**$H_{\text{phys}}$ : shellwise transport.** For every fixed shell, the scale spaces have compatible common-coordinate extensions, disjoint shell contours, stable rank, and summable Riesz step bounds. No global operator-norm convergence of the rank-growing cloud is requested.

## 3 Conditional physical-R closure theorem

**Theorem 1** (Shellwise closure of physical interface R). *Assume  $X_{\text{phys}}$ ,  $D_{\text{phys}}$ ,  $K_{\text{phys}}$ , and  $H_{\text{phys}}$ . Then for every fixed shell  $m$  and all sufficiently large scales  $j$ :*

1.  $\Gamma_{j,m}$  encloses a Riesz projection  $\Pi_{j,m}$  with  $\text{rank } \Pi_{j,m} = \text{rank } P_{j,m}$ ;
2.  $\text{ran } \Pi_{j,m}$  is a uniformly controlled graph over  $\text{ran } P_{j,m}$ , with a dual graph whenever requested by the marked-trace interface;
3.  $\Pi_{j,m}$  converges in common coordinates to a same-rank shell idempotent  $\Pi_{\infty,m}$ ;

4. *distinct limiting shells annihilate one another, so every finite partial cloud is a coherent Riesz projection of the expected rank.*

*This closes interface  $R$  in the rank-growing shellwise sense required by the moving-cloud determinant architecture.*

*Proof.* By  $X$  and the RH-162 realization theorem, the source packet becomes a type-correct transfer-space packet and its primal/adjoint defects bound the two directed couplings.  $D$  and RH-168 transfer validated nominal sample data to exact continuous block resolvent and coupling bounds.  $K$  activates the RH-163 rank homotopy and RH-166 directional graph theorems uniformly. Thus the first two conclusions hold at every sufficiently large scale.

For fixed  $m$ ,  $H$  activates the RH-169 summable common-coordinate transport, giving a same-rank norm limit. Disjoint finite-scale Riesz shells annihilate one another; continuity of multiplication preserves this in the limit. RH-170 then makes every finite partial sum an idempotent with additive rank. This is precisely the corrected shellwise  $R$  conclusion.  $\square$

**Remark 2** (Conditional means conditional). *Theorem 1 proves the implication from four named physical leaves. It supplies no evidence that any leaf holds. In particular, declaring an eigenvector fit to be  $J_{j,m}$  or treating sampled float inverses as outward bounds would merely assume  $X$  or  $D$ .*

## 4 Minimal frontier and omission mechanisms

Treat the displayed architecture as the monotone formula

$$R = X_{\text{phys}} \wedge D_{\text{phys}} \wedge K_{\text{phys}} \wedge H_{\text{phys}}.$$

**Proposition 3** (Architecture-relative completion frontier). *With current statuses, the unique inclusion-minimal missing bundle is*

$$\boxed{\{X_{\text{phys}}, D_{\text{phys}}, K_{\text{phys}}, H_{\text{phys}}\}}.$$

*Proof.* All four leaves are open, and the formula is a conjunction. Each is independent in the following proof-theoretic sense. Without  $X$ , the packet and transfer block are not on one space. Without  $D$ , a nonnormal eigenvalue gap does not bound the contour resolvent. Without  $K$ , a two-mode feedback can place spectrum on the contour even with exact finite data. Without  $H$ , independently certified scales can rotate or drift without a coherent shell limit. Thus deleting any leaf destroys the stated implication.  $\square$

These witnesses establish minimality only for this architecture. A future construction could bypass reset packets entirely and build Riesz shells directly; failure of  $X$  would reject the reset-to-cloud branch, not every possible determinant route.

## 5 Aggregate finite audit

The reproducibility script reads the archived outputs of RH-162–170. The matrix audits comprise:

| layer group                 | finite cases | recorded failures |
|-----------------------------|--------------|-------------------|
| ambient, Schur, balance     | 1,536        | 0                 |
| midgap and bi-Ritz          | 1,024        | 0                 |
| mesh and operator-ball      | 512          | 0                 |
| common-coordinate transport | 512          | 0                 |
| rank-change witnesses       | 63           | 0                 |

The total is 3,584 finite matrix cases plus 63 exact rank-change witnesses. The audits test formulas and numerical implementations. They do not sample the missing physical realization or prove an asymptotic law.

## 6 Updated Gate-A coordinate

The complete Gate-A dependency after RH-171 is

$$(S_{\text{native}} \text{ or } S_{\text{lagged}}) \wedge R \wedge Q \wedge U \wedge Z \wedge T.$$

RH-171 refines only

$$R = X_{\text{phys}} \wedge D_{\text{phys}} \wedge K_{\text{phys}} \wedge H_{\text{phys}}.$$

The reset seed  $S$  remains conditional;  $Q$ ,  $U$ ,  $Z$ , and  $T$  remain open. Therefore macro Gate A is not closed and Gate B has not begun in the strict dependency order.

The recommended next target is  $X_{\text{phys}}$ . Until a canonical ambient map exists, transfer-space Ritz residuals and contours attached to the reset packet are not type-defined. If  $X$  survives, the validated finite pipeline makes  $D$  the next concrete wall. If  $X$  fails, that negative result cleanly rejects this branch while preserving direct-cloud alternatives.

No canonical all-level determinant, scattering completion, self-adjoint operator,  $T \log T$  law, von Mangoldt trace formula, zeta divisor identity, or Riemann Hypothesis is claimed.

## References

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